

Why we ran this

Directing generative video from a 3D scene

Not to replace the cinematographer. To give them an editable tool.

Experimental material · current possibilities and limits of the workflow



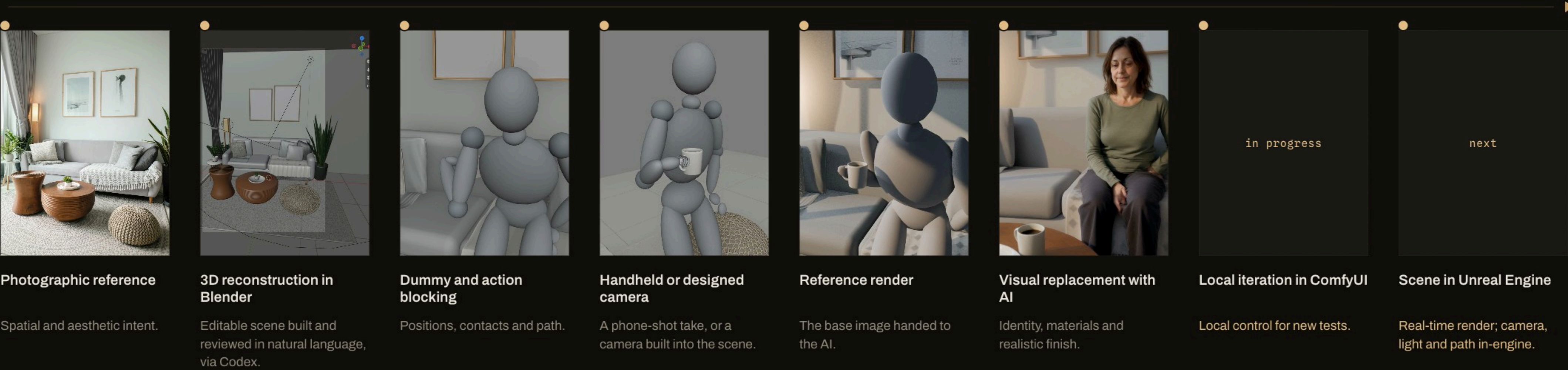
Reference · real living room
environment.jpg

Magnific · visual replacement
final result of the experiment



Character reference
character_reference_woman_40.png

The pipeline



The same scene: reference photo and rebuilt 3D



Photographic reference
environment.jpg



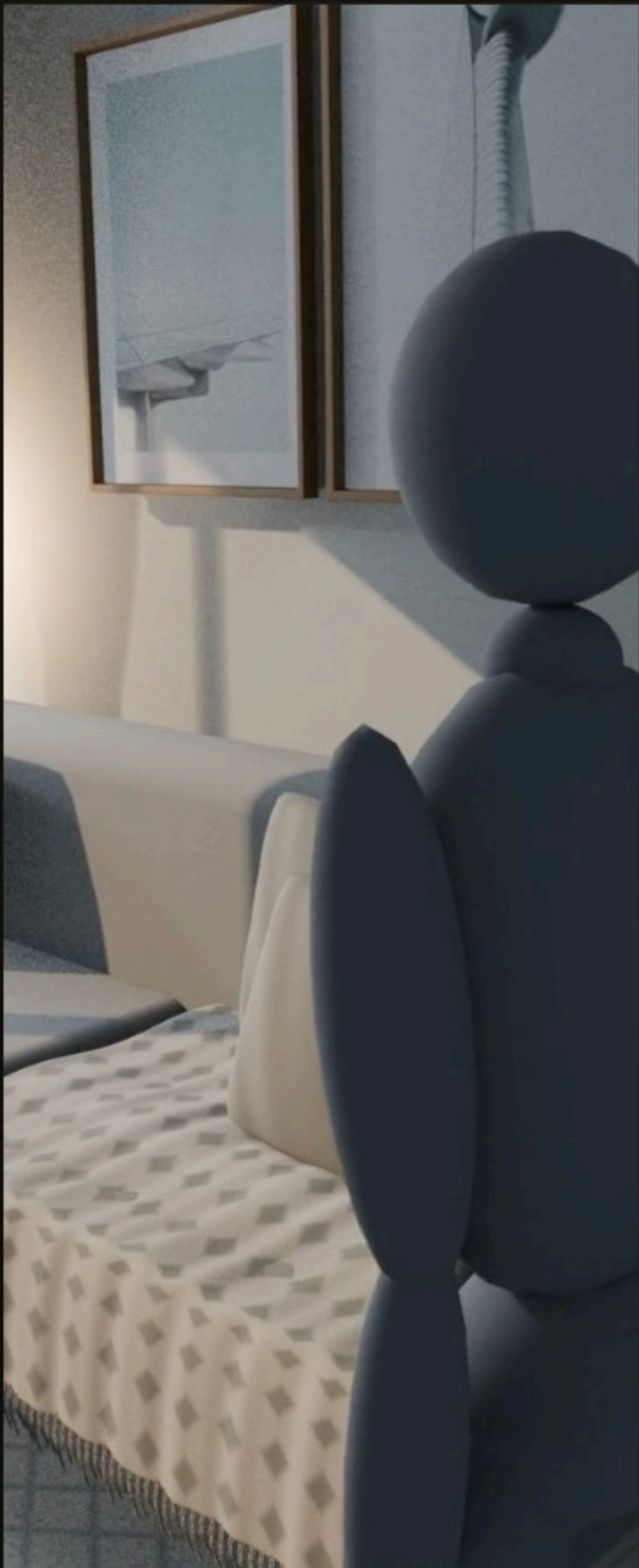
3D reconstruction in Blender
automatic render of the scene



Blender + MCP · port 9876

From the photo, Codex rebuilt the living room in Blender over an MCP connection: geometry, scale, materials, camera and light, in natural language. The scene stays editable and gets fixed without leaving the workflow.

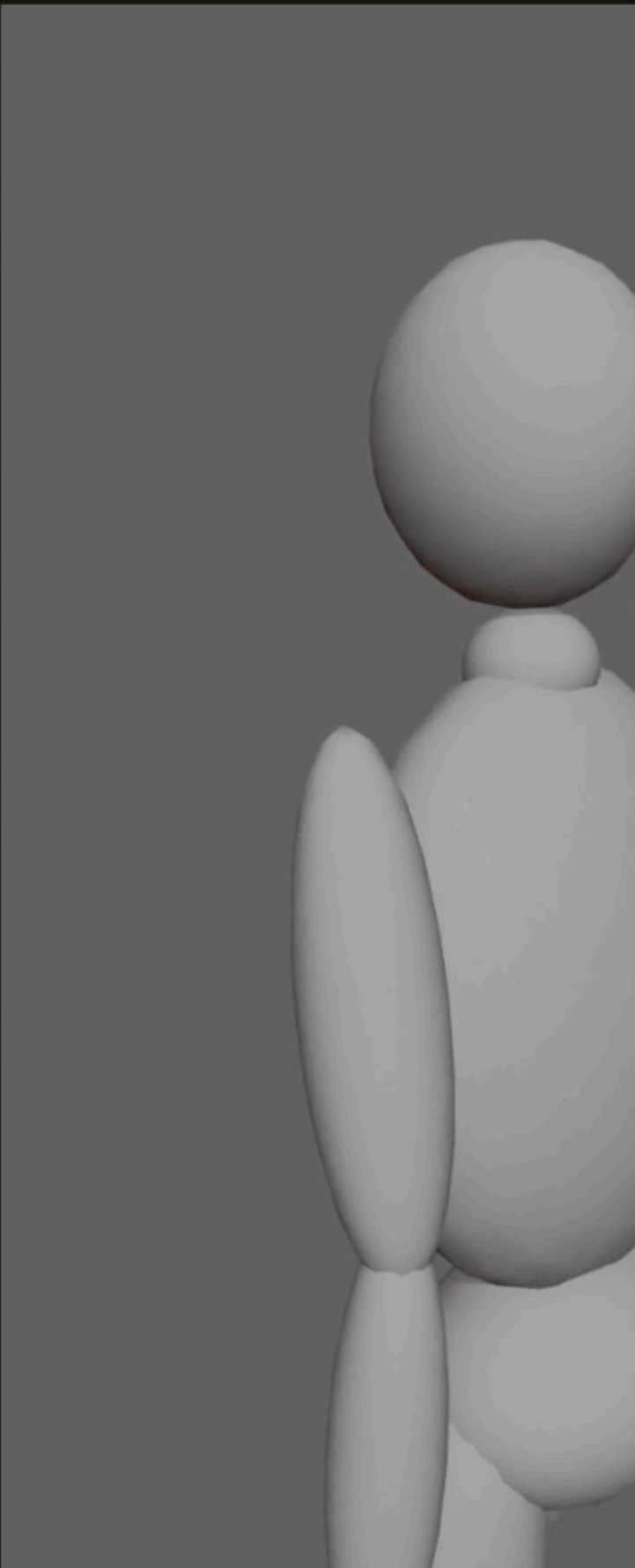
Blender as a control space



Blocking · mid-afternoon coffee
cafe_media_tarde_blocking.mp4



Environment on its own
test_manual_entorno_completo.mp4



Simplified dummy
test_manual_dummy_brazo_suave.mp4

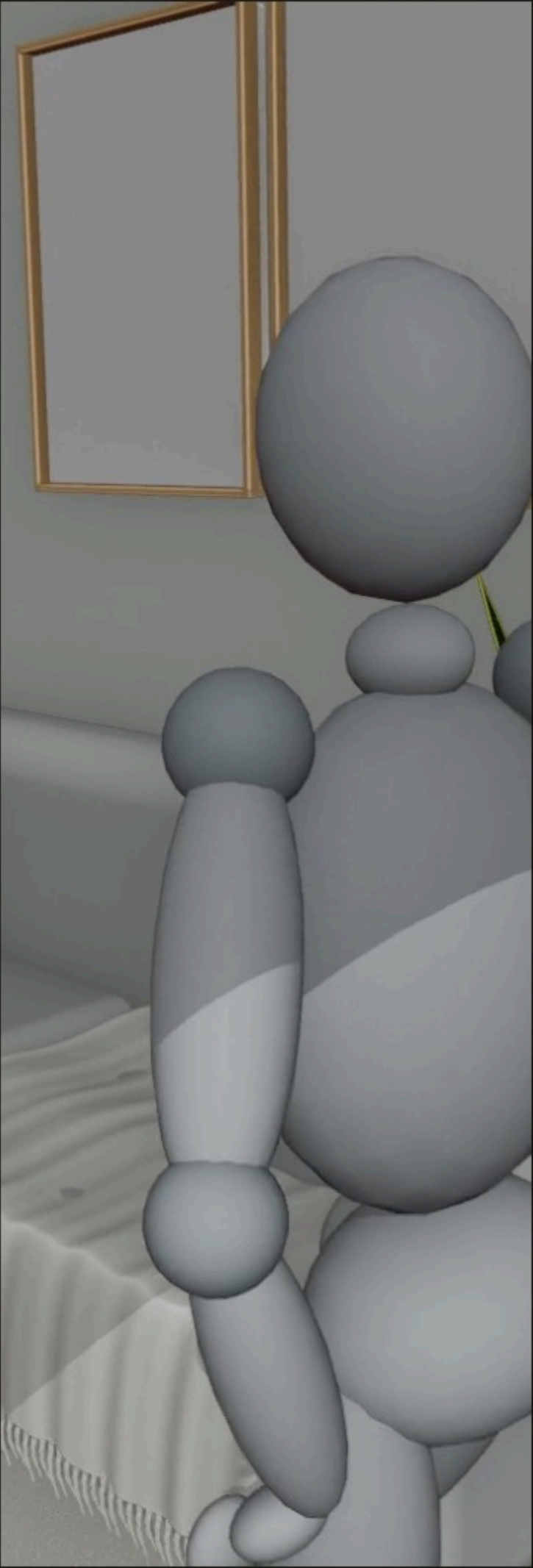
Blender keeps what a video generator usually can't offer:
editable geometry, scale, paths, positions, light and camera.

Codex automated building the living room, the materials, the dummy, the cameras, the renders and the fixes.

Camera as a human gesture

VirtuCamera turned the phone into a virtual camera. The take could be recorded by hand inside the scene, like a camera operated by a person.

- The phone defines the camera gesture.
- Blender keeps the path and lets you repeat it.
- The camera doesn't have to be generated from scratch in Magnific.



Handheld take · VirtuCamera
test_manual_1.mp4



Camera designed in the scene
coffee_advert_camera.mp4

What worked and what held the result back

Worked

- + Fast reconstruction of an editable scene.
- + Control over camera and composition.
- + Character and environment references with a consistent look.
- + Iteration between 3D and video generation.

Held back

- The generated video inherited some of the dummy's stiffness.
- Prompts don't fix an unnatural body performance.
- Hands, contacts and weight shifts are still fragile.
- Visual replacement doesn't stand in for a convincing motion base.



Magnific · final result
magnific_transform...1lPTYjJr4r.mp4

From the Magnific render to local control



First test · Magnific
magnific_use-vid1-as-the-source...VXaaflIMMU.mp4



Final render · Magnific
magnific_transform...1lPTYjJr4r.mp4



Blocking in Blender
source scene

We already have a final render in Magnific. It confirms the pipeline can carry character and environment into a realistic image.

- Now
- ComfyUI: iterate locally, more control over each test and less dependence on credits.
- Next
- Take the 3D scene into Unreal Engine: real-time render, with camera, light and path inside the engine.

3D works as a shared language between directing, camera, animation and video generation.

Credits

Miguel Ángel Ballesteros

Alba Díaz

Teyo Cuadrado

An experiment with Blender, Codex and generative video.